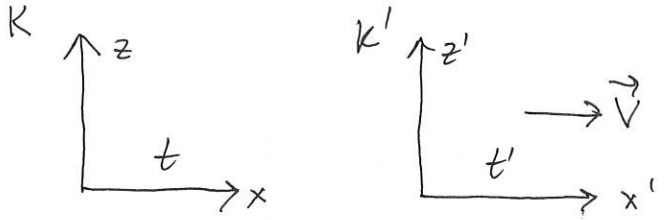
# Special relativity.

Inertial frame = Uniformly moving observer with a clock and a meter stick.




Newtonian mechanics ①

$$\left.\begin{aligned} \vec{r}_i &= \vec{r}_i' + \vec{V}t \\ t &= t' \end{aligned}\right\} \quad \frac{d\vec{r}_i}{dt} = \frac{d\vec{r}_i'}{dt'} + \vec{V}, \quad \frac{d^2\vec{r}_i}{dt^2} = \frac{d^2\vec{r}_i'}{dt'^2}$$

$$\vec{v}_i = \vec{v}_i' + \vec{V}, \quad \vec{a}_i = \vec{a}_i'$$

2nd law

$$\Rightarrow m_i \frac{d^2\vec{r}_i}{dt^2} = \sum_{j\neq i} \vec{f}_{ij} = -\vec{\nabla}_i \sum_{j\neq i} U(|\vec{r}_i - \vec{r}_j|)$$

force from particle j on particle i

Under ~~these~~ Galilean transformation ① $\vec{r}_i - \vec{r}_j = \vec{r}_i' - \vec{r}_j'$

$$\vec{a}_i = \vec{a}_i'$$

$\Rightarrow$ Newton's 2nd law is invariant.

② Electromagnetism

Lorentz force $\vec{F}_L = q\vec{E} + q\vec{v}\times\vec{B}$

But $\vec{v}$ depends on reference frame $\Rightarrow \vec{F}_L$ is not Galilean-invariant.

Also
$$\begin{cases} \vec{\nabla}\times\vec{E} = -\dfrac{\partial\vec{B}}{\partial t} \\ \vec{\nabla}\times\vec{B} = \dfrac{1}{c^2}\dfrac{\partial\vec{E}}{\partial t} \end{cases}$$

$c = 2.998\cdot 10^8$ m/s speed of light

$\Rightarrow$ Q: on which frame this c is defined?

A: [ether theory] there exist a special reference frame, invisible "ether" where c is defined.
1 Maxwell...

$A_2$: [Poincare, Lorentz, Einstein] All inertial frames are equivalent and Maxwell equations look the same in all of them.
1898 1904 1905

$\Rightarrow$ Speed of light $c$ is <u>the same</u> in all inertial frames!

$\Rightarrow$ this is <u>principle of relativity</u>

1) laws of physics are the same in all ref. frames
2) speed of light is the same: $c = c'$.

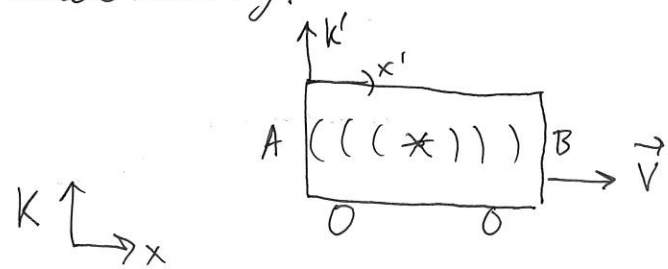
Immediate consequence: time is <u>not absolute</u>, $t \neq t'$.

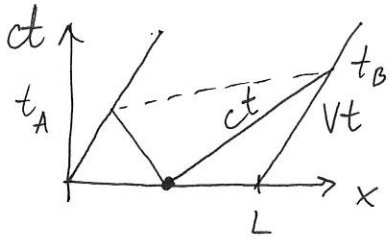
4 Thought experiments

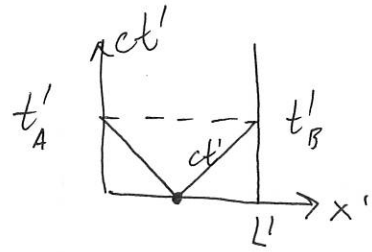
(a) Trains move in opposite directions. Someone is drawing a chalk line on each at the same height. Are the heights seen by observers on the trains the same? — Yes, by symmetry.

$\Rightarrow$ perpendicular to the motion directions do not transform, $\vec{x}'_\perp = \vec{x}_\perp$.

(b) Simultaneity.



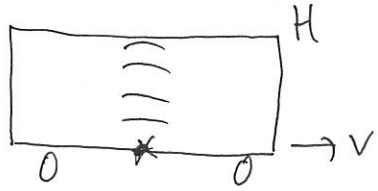



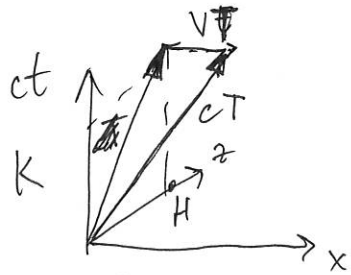

space-time diagram.

Two events simultaneous in $K'$ ($t'_A = t'_B$) are not simultaneous in $K$ ($t_A \neq t_B$).

(c) Time dilation




Flash a light on the floor of the car, it strikes the ceiling at height H.




Light went up and to the right vT.

$H' = H$ from (a).

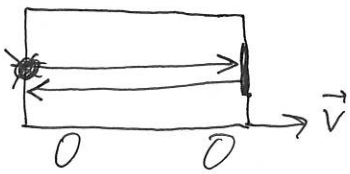
$\Rightarrow$ in $K'$ : $H' = H = cT'$

in $K$ : $cT = \sqrt{H^2 + v^2T^2} = \sqrt{c^2T'^2 + v^2T^2}$

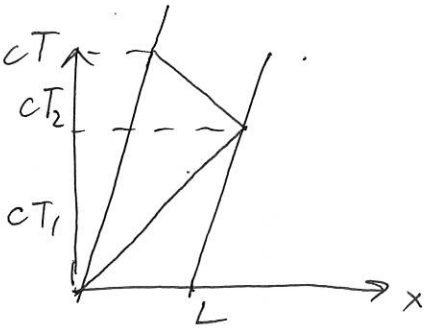
$$c^2T^2 = c^2T'^2 + v^2T^2 \Rightarrow T = \frac{c}{\sqrt{c^2-v^2}}\,T' = \frac{T'}{\sqrt{1-v^2/c^2}} = \gamma T'$$

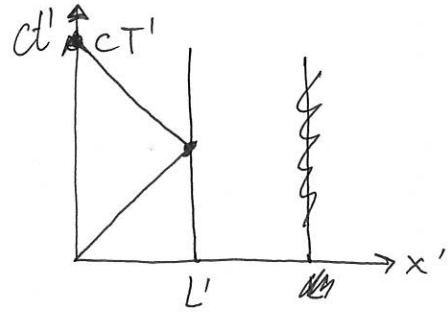
$$\frac{T}{T'} = \gamma = \frac{1}{\sqrt{1-v^2/c^2}} \geq 1 \Rightarrow$$ moving clock runs slow. Time dilation.

(d) length contraction.




Flash light on one wall, it reflects off a mirror on the opposite and returns.




$$cT_1 = L + vT_1, \quad T_1 = \frac{L}{c-v} \qquad\qquad cT' = 2L'$$

$$cT_2 = L - vT_2, \quad T_2 = \frac{L}{c+v}$$

$$T = T_1 + T_2 = \frac{2cL}{c^2-v^2} = \gamma T' = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}\,\frac{2L'}{c}$$

$$cL = \frac{c^2-v^2}{\sqrt{1-\frac{v^2}{c^2}}}\,\frac{L'}{c} = \frac{\sqrt{c^2-v^2}}{\phantom{c}}\,L'$$

$$\frac{L}{L'} = \sqrt{1-\frac{v^2}{c^2}} = \frac{1}{\gamma} \Rightarrow L = \frac{L'}{\gamma} < L'$$

$\Rightarrow$ length contraction

# The Lorentz transformation

Exp (a) says that $y = y'$, $z = z'$.

Exp (b) that $x$ and $t$ are mixed.

We define $x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$.

Exp (c): if $x^{1'} = 0$, then $\frac{x^0}{x^{0'}} = \gamma$.

The origin of the moving frame $x^{1'} = 0$ moves according to $x^1 = vt = \frac{v}{c} x^0 = \gamma \beta x^{0'}$

where $\beta \equiv \frac{v}{c}$, $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$

All together:

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \gamma x^{0'} + \gamma A(\beta) x^{1'} \\ \gamma \beta x^{0'} + \gamma B(\beta) x^{1'} \end{pmatrix} = \Lambda(\beta) \begin{pmatrix} x^{0'} \\ x^{1'} \end{pmatrix}$$

$$\Lambda(\beta) = \gamma \begin{pmatrix} 1 & A(\beta) \\ \beta & B(\beta) \end{pmatrix}$$ with $A, B$ not determined

But we can un-do the transformation by $\Lambda(-\beta)$ so that $\Lambda(\beta)\Lambda(-\beta) = \mathbb{1}$

$$\Rightarrow \gamma \begin{pmatrix} 1 & A(\beta) \\ \beta & B(\beta) \end{pmatrix} \gamma \begin{pmatrix} 1 & A(-\beta) \\ -\beta & B(-\beta) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\gamma^2 \begin{pmatrix} 1 - \beta A(\beta) & A(-\beta) + A(\beta) B(-\beta) \\ \beta - \beta B(\beta) & \beta A(-\beta) + B(\beta) B(-\beta) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Thus $\quad 1-\beta A(\beta) = \frac{1}{\gamma^2} = 1-\beta^2 \Rightarrow A(\beta)=\beta.$

$$\beta - \beta B(\beta) = 0 \Rightarrow B(\beta) = 1.$$

Then $\quad A(-\beta) + A(\beta)B(-\beta) = -\beta + \beta\cdot 1 = 0.$

$$\beta A(-\beta) + B(\beta)B(-\beta) = \beta(-\beta) + 1 = \frac{1}{\gamma^2} = 1-\beta^2. \quad \text{Ok.}$$

$$\Rightarrow \Lambda(\beta) = \gamma\begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}, \quad A(-\beta) = \gamma\begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix}$$

$$\Lambda(\beta) = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x^{0'} \\ x^{1'} \\ x^{2'} \\ x^{3'} \end{pmatrix}$$

Definition: $\quad x^i = (ct, \vec{x})$ contravariant 4-vector

$$x^{\alpha'} = \Lambda^\alpha{}_\mu(\vec{\beta})\, x^\mu$$

$$x^\alpha = \Lambda^\alpha{}_\mu(-\vec{\beta})\, x^{\mu'}$$

Position of indices is important. Will become clear later.

$$\Lambda^\alpha{}_\nu(-\beta)\,\Lambda^\nu{}_\mu(\beta) = \delta^\alpha_\mu = \begin{cases} 1, & \alpha=\mu \\ 0, & \alpha\neq\mu \end{cases}$$

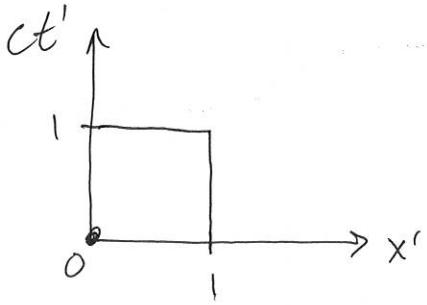
Another succinct form:

$$x^0 = \gamma\left(x^{0\prime} + \vec{\beta}\cdot \vec{x}_{\parallel}^{\,\prime}\right)$$

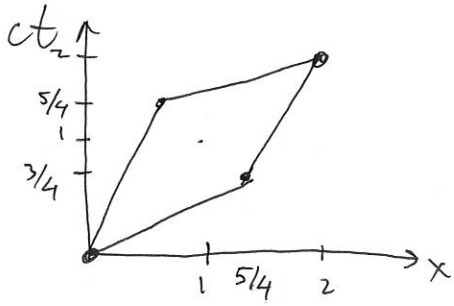
$$\vec{x}_{\parallel} = \gamma\left(\vec{x}_{\parallel}^{\,\prime} + \vec{\beta}\, x^{0\prime}\right)$$

$$\vec{x}_{\perp} = \vec{x}_{\perp}^{\,\prime}.$$

let $\beta = 3/5$, $\gamma = \dfrac{1}{\sqrt{1-\left(\frac{3}{5}\right)^2}} = \dfrac{1}{\sqrt{\frac{16}{25}}} = \dfrac{5}{4}$. $\Rightarrow \gamma\beta = \dfrac{3}{4}$.




Events $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$.




$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \frac{5}{4}\begin{pmatrix} 1 & \frac{3}{5} \\ \frac{3}{5} & 1 \end{pmatrix}\begin{pmatrix} x^{0\prime} \\ x^{1\prime} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{5}{4} \end{pmatrix}\begin{pmatrix} x^{0\prime} \\ x^{1\prime} \end{pmatrix}$$

The interval: Lorentz transformation is designed to preserve the speed of light

$\Rightarrow$ equation for the propagation of light

$$x = ct$$

or, $$c^2 t^2 = \vec{x} \cdot \vec{x}$$

$$c^2 t^2 - \vec{x} \cdot \vec{x} = 0$$

is invariant.

$$(x^0)^2 - (x^1)^2 = (\gamma x^{0'} + \gamma\beta x^{1'})^2 - (\gamma\beta x^{0'} + \gamma x^{1'})^2 =$$

$$= \gamma^2 (1-\beta^2)(x^{0'})^2 + \gamma^2(\beta^2 - 1)(x^{1'})^2$$

$$= \frac{1-\beta^2}{1-\beta^2}\left[(x^{0'})^2 - (x^{1'})^2\right] = (x^{0'})^2 - (x^{1'})^2.$$

$$c^2 t^2 - \vec{x} \cdot \vec{x} = x_j x^j = x^j x_j = x^i g_{ij} x^j$$

Covariant 4-vector $x_j = (ct, -\vec{x})$

Metric tensor $g = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$

$x_j = x^i g_{ij}$ where summation over repeated indices ($i$ in this case) is assumed.

The interval = how far, in space and time, two events occur

$$s_{12}^2 = c^2(t_1 - t_2)^2 - (\vec{r}_1 - \vec{r}_2)^2$$

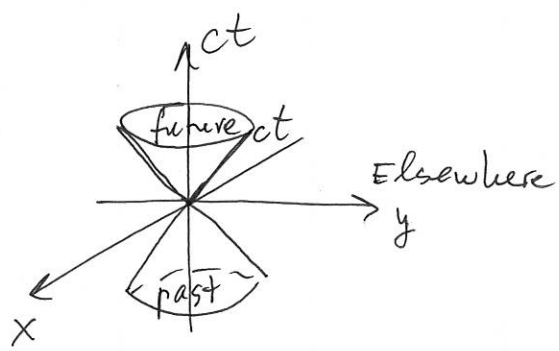
Event: collision of two particles, emission/absorption of light...

For light signal $\vec{r}^{\,2} = c^2 t^2 \Rightarrow s_{12} = 0$

The interval is invariant $\Rightarrow s_{12} = 0 \longrightarrow s'_{12} = 0$.

$s_{12}^2 > 0 \Rightarrow$ timelike interval } causal relationship
$s_{12}^2 = 0 \Rightarrow$ light line interval } here

$s_{12}^2 < 0 \Rightarrow$ spacelike interval : no cause here since spatial distance $r > ct$ here.



Velocity transformations

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$d(ct) = \gamma\, d(ct') + \gamma\beta\, dx'$$
$$dx = \gamma\beta\, d(ct') + \gamma\, dx'$$

$$dy = dy', \quad dz = dz'$$

$$v_x = \frac{dx}{dt} = c\,\frac{dx' + \beta c\, dt'}{\beta\, dx' + c\, dt'} = c\,\frac{\left(\frac{dx'}{dt'} + \beta c\right) dt'}{c\, dt'\left(1 + \frac{\beta}{c}\frac{dx'}{dt'}\right)} =$$

$$\left(\beta = \frac{v}{c}\right) \qquad = \frac{v_x' + v}{1 + \frac{v_x' v}{c^2}}$$

Recall Galilean result: $v_x = v_x' + v$

The difference become significant when $v_x', v \sim c$.

Suppose $v = \frac{3c}{4}$ and $v_x' = \frac{3c}{4}$

then $v_x\big|_{\text{Galilean}} = \frac{3c}{4} + \frac{3c}{4} = \frac{3}{2}c$

while $v_x\big|_{\text{Lorentz}} = \dfrac{\frac{3c}{4} + \frac{3c}{4}}{1 + \left(\frac{3c}{4c}\right)^2} = \dfrac{\frac{3}{2}c}{1 + \frac{9}{16}} = \frac{16}{25}\cdot\frac{3c}{2} =$

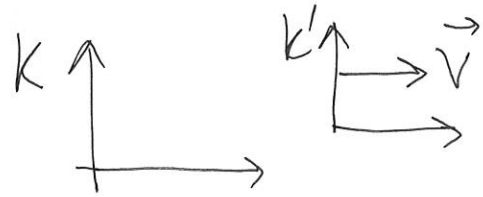
$$= \frac{24}{25}c.$$

$$v_y = \frac{dy}{dt} = \frac{c\, dy'}{\gamma(\beta\, dx' + c\, dt')} =$$

$$= \frac{c\, dy'}{\gamma c\, dt'\left(1 + \frac{\beta}{c}\frac{dx'}{dt'}\right)} = \frac{v_y'}{\gamma\left(1 + \frac{v_x' v}{c^2}\right)}$$

Proper time

Clock in $K'$, which moves with $\vec{v}$ w.r.t. $K$




$$K': ds^2 = c^2 dt'^2 - (d\vec{r}')^2 = c^2 (dt')^2$$

$$K: ds^2 = c^2 (dt)^2 - (d\vec{r})^2 =$$
$$= c^2 (dt)^2 - (v\,dt)^2 =$$
$$= (c^2 - v^2)(dt)^2$$

$$\Rightarrow dt' = \sqrt{1 - \frac{v^2}{c^2}}\, dt = \frac{dt}{\gamma} \leq dt$$

Proper time $d\tau = \dfrac{dt}{\gamma}$